Sales Forecasting & Performance Dashboard

Using Machine Learning

Dataset: Superstore Sales (Kaggle)

Tools: Python, Scikit-learn, Pandas, Matplotlib, Excel

Records: 9,994 transactions

1. Problem Statement

The objective of this project is to analyze retail sales data from the Superstore dataset and build machine learning models to predict future sales. This involves data cleaning, exploratory data analysis (EDA), feature engineering, model training, evaluation, and creating an interactive Excel dashboard for business stakeholders.

2. Dataset Description

Source: Kaggle - Superstore Sales Dataset
Total Records: 9,994
Features: 25 columns including Order Date, Ship Mode, Segment, Category, Sub-Category, Sales, Quantity, Discount, Profit, Region
Date Range: 2014-01-03 to 2017-12-30
Total Sales: $2,297,200.86
Total Profit: $286,397.02

3. Methodology

Step 1 - Data Cleaning: Handled missing values, removed duplicates, parsed dates, and engineered time-based features (Year, Month, Day of Week, Shipping Days).

Step 2 - Exploratory Data Analysis: Analyzed sales distribution by category, region, segment, and sub-category. Examined monthly trends and discount-profit relationships.

Step 3 - Feature Engineering: Encoded categorical variables (Ship Mode, Segment, Region, Category, Sub-Category) using Label Encoding. Selected 11 features for modeling.

Step 4 - Model Training: Trained two regression models:
- Linear Regression (baseline)
- Random Forest Regressor (ensemble method)

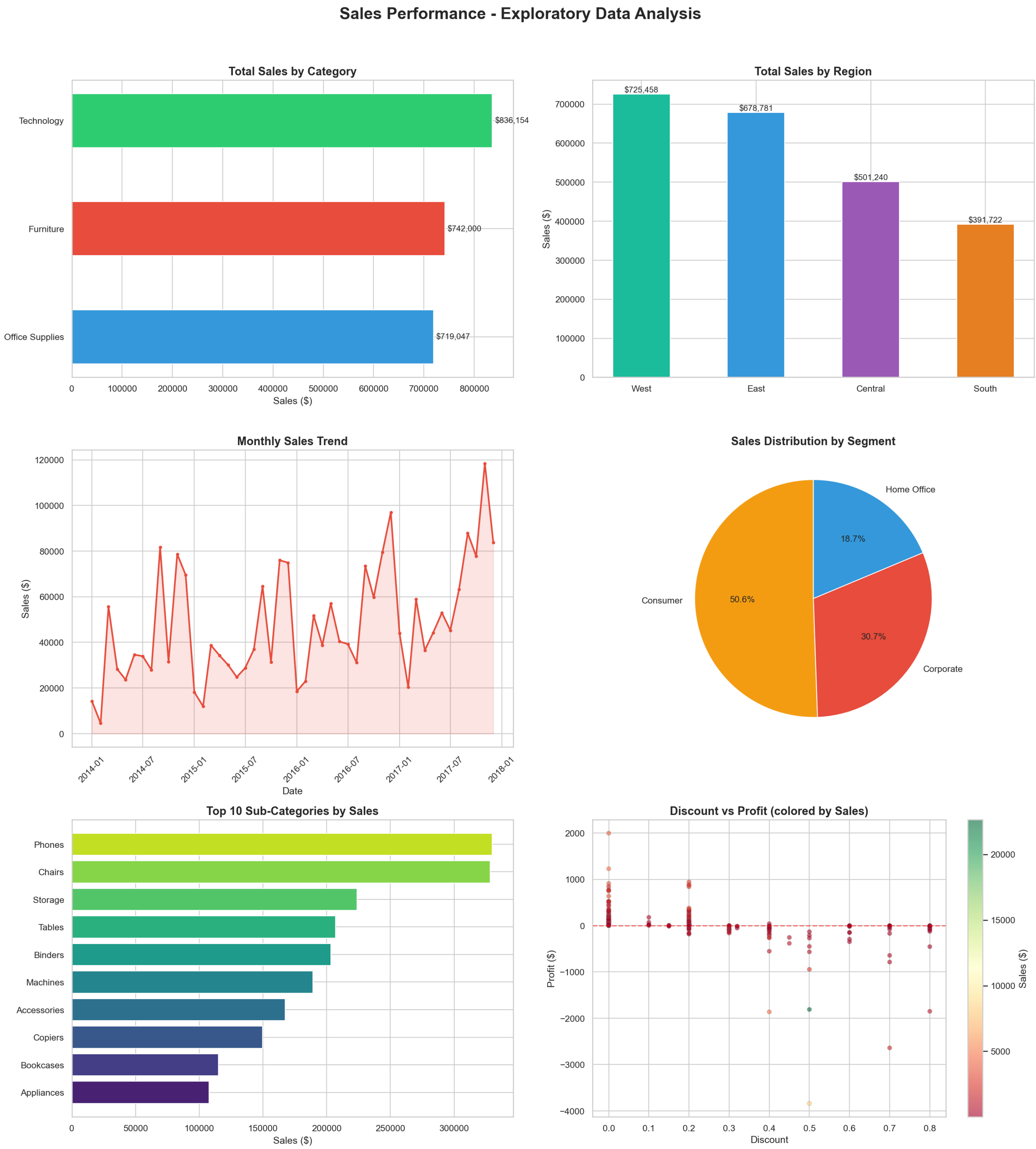
Step 5 - Evaluation: Compared models using R-squared Score, MAE, and RMSE.

4. Model Results

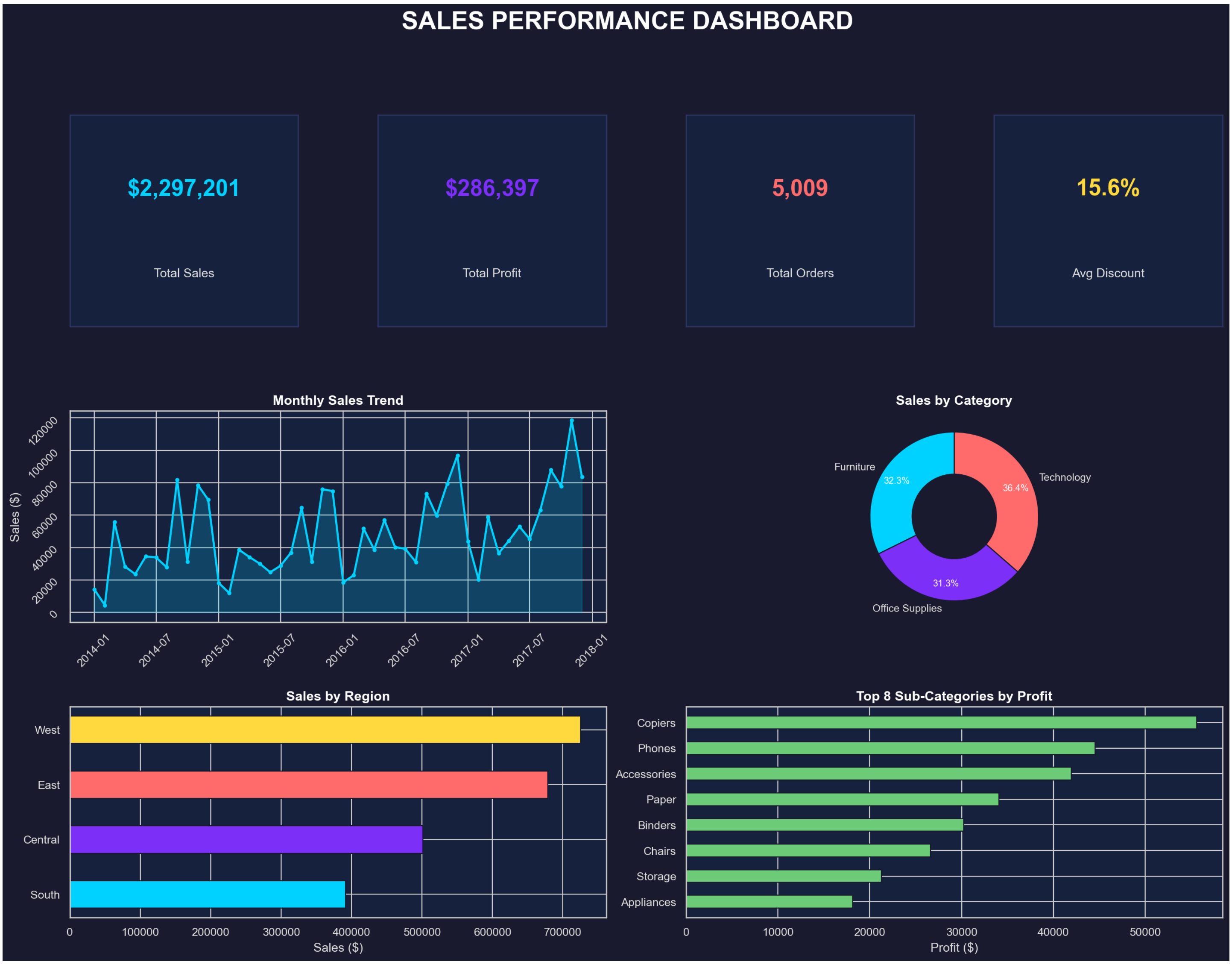
Model	R2 Score	MAE	RMSE
Linear Regression	0.0438	273.40	751.56
Random Forest	0.1811	220.10	695.52

Best Model: Random Forest with R2 Score of 0.1811

5. Exploratory Data Analysis - Charts



6. Dashboard Preview



7. Conclusion

The Random Forest model achieved the best performance with an R2 score of 0.1811, MAE of 220.10, and RMSE of 695.52. Key business insights include:

- Technology category generates the highest sales revenue
- Consumer segment is the largest customer base
- West region leads in total sales
- Higher discounts tend to reduce profit margins
- Sales show seasonal patterns with peaks in Q4

The Excel dashboard provides an interactive view for stakeholders to explore sales performance across different dimensions.